

Class C NPD: GMAC Class C Sweep Report

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Setup

- **Channel:** Gaussian MAC, SNR = 6 dB
- **Path:** Class C (`path_i` = N), decode all U first then V
- **Rate:** 50% of per-user capacity ($R_u \sim 0.232$, $R_v \sim 0.456$)
- **Architecture:** single-user NPD per stage, $d=16$ hidden=64 (~21K params)
- **Training:** fast_ce parallel, curriculum across N with warm-starting
- **Evaluation:** 5000 codewords per N, Wilson 95% CIs

SC Reference

N	ku	kv	SC BLER	NPD target (1.5x)
16	4	7	0.1626	0.2439
32	7	15	0.0684	0.1026
64	15	29	0.0266	0.0399
128	30	58	0.0054	0.0081
256	59	117	0.0020	0.0030
512	119	233	0.0008	0.0012
1024	238	467	0.0002	0.0003

Curriculum Sweep Results

N	Stage 1 BLER	Stage 2 BLER	Chained BLER	95% CI	Ratio	Pass?
16	0.1420	0.0060	0.1362	[0.1270, 0.1460]	0.84x	[OK] PASS
32	0.1160	0.0000	0.1290	[0.1200, 0.1386]	1.89x	[NO] FAIL
64	0.1600	0.0000	0.1758	[0.1655, 0.1866]	6.61x	[NO] FAIL

Summary

- **PASSED** at: 16
- **FAILED** at: 32, 64

Architecture diagram

GMAC channel: $z = (1-2X) + (1-2Y) + W$

Stage 1 (NPD) U on marginal channel
decode U from raw z mixture LLR via `z_encoder`

$\hat{u}$

$\hat{x} = \text{polar_encode}(\hat{u})$

$z' = z - (1 - 2 \cdot \hat{x})$ subtract known U contribution

Stage 2 (NPD) V on clean BPSK+AWGN
decode V from z' standard single-user NPD

$\hat{v}$

output

Reproduction

Run smoke test (5 min)

python -u class_c_npd/smoke_test.py

Run curriculum sweep

python -u class_c_npd/training/curriculum_sweep.py --N_list 16,32,64,128

Generate this report

python -u class_c_npd/eval/generate_report.py